

LIMS Print Pipeline

High-level overview — how a printed lab document travels from an offline client PC to the cloud LIMS

The setup

60 lab client PCs have **no internet** — only the local network. One central desktop (**192.168.1.172**) has internet and hosts a shared folder, **limsDocs**, that every client can reach. This toolkit turns each client into a **virtual printer**: staff print from any application, and the document is captured as a PDF, tagged with a registration number and test, filed into the shared folder, and pushed live to a cloud **LIMS** web app — all without the clients ever touching the internet.

How it flows

60 Client PCs (offline, LAN only)

Staff print from any app → a dialog offers Registration → Test dropdowns
The job becomes a PDF (optionally AES-256 encrypted) and is POSTed to the hub

print → pick registration → pick test

Central Hub · 192.168.1.172:8000 (has internet, hosts limsDocs)

Files each PDF into `limsDocs\<device>\<type>\<registration>\<test>\`
Holds any job missing a valid registration · syncs the catalog from the cloud (~2s)

file to share + forward instantly

Cloud LIMS · Supabase + React

Testers create registration numbers (device type + tests)
Printed documents appear live; download any PDF from anywhere

The four building blocks

1 Client printer

A normal Windows printer. Converts print jobs to PDF and asks the user to pick the registration & test from live dropdowns. Per-printer password, device name and type.

2 Central hub

The one internet-connected desktop. Receives every job, files it into the `limsDocs` tree, holds anything unregistered, and forwards each PDF to the cloud with automatic retry.

3 Cloud LIMS (web app)

React + Supabase. Testers create registration numbers with their device type and tests; printed documents stream in live and are downloadable.

4 Registration catalog

The link between them: registrations created in the cloud sync to the hub every ~2 seconds and drive the client dropdowns — so numbers are never typed by hand.

What one print looks like

- 1 Print** Staff member prints from Word, Chrome, an instrument app — anything.
- 2 Choose** A dialog shows the open registrations for that printer's device type; pick one, then its test.
- 3 Convert** The job is turned into a PDF (optionally AES-256 encrypted) and sent to the hub.
- 4 File** The hub saves it to `limsDocs\device\type\registration\test\` — or holds it if the number is missing.
- 5 Sync** The PDF is forwarded to the cloud LIMS and appears in the Documents page within seconds.

Nothing is ever lost

A registration number is **required**. If a user skips it, the number doesn't match, or the hub is briefly unreachable, the job is never dropped: the hub **holds** it (safely stored, not in the share) until an operator assigns the right registration in the dashboard, and the cloud forward **retries automatically** until it succeeds. Upload failures on the client are preserved locally too.

Built for security

- **Per-printer tokens.** Each printer has its own secret ingest token; the hub decides the true device identity.
- **Passwords protected.** PDF passwords are encrypted at rest (machine-bound DPAPI); viewing one needs the super-admin password.
- **Cloud key stays home.** The Supabase service key lives only on the central desktop; the web app uses a limited key + row-level security.
- **No path escapes.** Every folder name from a client is sanitized — a malicious document name can't escape the limsDocs tree.
- **Encryption fails closed.** If encryption is on but the password is missing, the job is held back rather than sent unprotected.
- **Locked install.** The client install folder is restricted to SYSTEM + Administrators, closing local privilege-escalation paths.

Deploying it (four moves)

- 1 **Cloud.** Create a Supabase project, run the provided schema, add tester logins. Start the React web app.
- 2 **Hub.** On 192.168.1.172, run the hub; note its admin token & enroll key; point it at the Supabase project and the limsDocs folder.
- 3 **Clients.** On each of the 60 PCs, run install — give it a unique device name, device type, PDF password and the enroll key.
- 4 **Go live.** Testers create registrations in the web app; staff print and pick them from dropdowns. Documents flow to the cloud.

Full step-by-step instructions live in **LAN-SETUP.md** in the project.

Verified end-to-end

The whole chain was tested with real components: real print → PDF conversion → hub filing → hold → assign → cloud forwarding, plus catalog sync, retry recovery, password encryption and the print dialog. 29/29 hub checks and 7/7 dialog checks passed.